RAG Learning

P6 — Global Challenges

OCR GCSE Physics A (J249) — Gateway Science · Paper 2

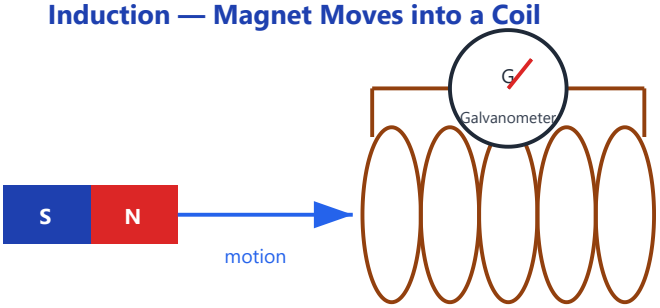
P6.1 Physics on the Move — Vehicles, Stopping, Safety

This section draws heavily on material from P2 (Forces). You should be able to apply Newton's laws and momentum to real-world transport situations, including safety features and stopping distances.

Electromagnetic Induction

> When a **conductor moves through a magnetic field**, or a **magnetic field changes around a conductor**, a potential difference is **induced** across the conductor. If the conductor is part of a complete circuit, a current flows. This is **electromagnetic induction** — the basis of *all* modern generation of electricity.

Induction — Magnet Moves into a Coil



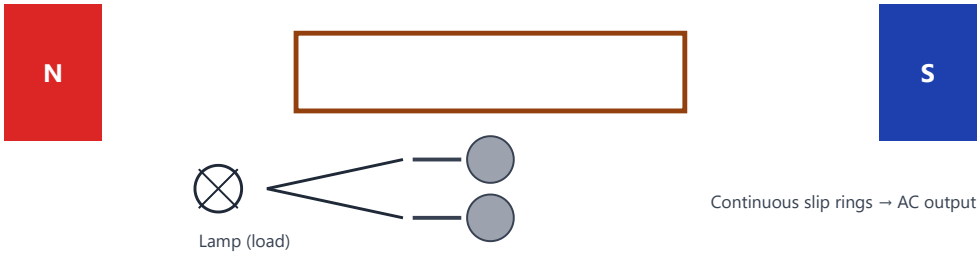
As magnet moves: flux through coil changes → pd induced → current → meter deflects.
Stop moving the magnet → no change of flux → no induced pd → no current.

How Generators Work

A **generator** is essentially a motor run in reverse. A coil is rotated in a magnetic field — the flux through the coil constantly changes, inducing an alternating pd that produces an **alternating**

current. Power stations all use generators; what differs is how the coil is turned (steam from a boiler, water from a dam, wind, etc.).

Simple AC Generator



Transformers & National Grid Recap

A **transformer** uses two coils on an iron core. Alternating current in the primary coil creates a changing magnetic field in the core, which induces an alternating pd in the secondary coil.

$$\mathbf{V_p / V_s = N_p / N_s} \quad \text{(ratio of primary/secondary voltages = ratio of turns)}$$

For an ideal transformer: $\mathbf{V_p I_p = V_s I_s}$ (power in = power out).

Worked Example: A transformer has 2 000 turns on the primary and 100 on the secondary. 230 V ac is applied to the primary. Secondary voltage?

$V_s = V_p \times N_s / N_p = 230 \times 100 / 2000 =$ **11.5 V**.

It's a **step-down** transformer (fewer turns on secondary).

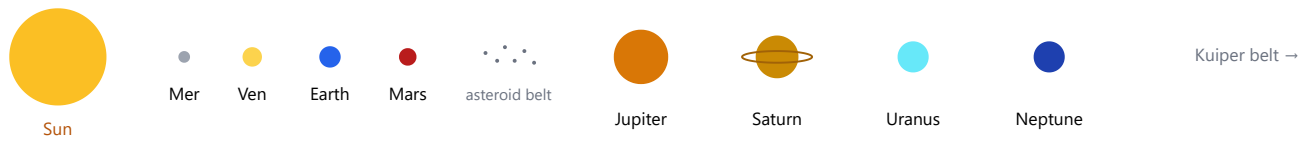
Uses: step-up at power stations (e.g. 25 kV → 400 kV) for efficient long-distance transmission; step-down near homes (400 kV → 11 kV → 230 V) for safe use.

P6.2 The Solar System

Our **Solar System** has one star (the Sun) and eight planets orbiting it, along with dwarf planets (e.g. Pluto), moons, asteroids and comets. It formed about 4.6 billion years ago from a cloud of

> gas and dust (a **nebula**).

The Solar System (not to scale)



Type	Examples	Orbit
Star	The Sun	Centre of Solar System
Planets	Mercury → Neptune (8)	Orbit the Sun in near-circular ellipses
Dwarf planets	Pluto, Ceres, Eris	Orbit Sun, but not cleared their orbital path
Moons / natural satellites	Earth's Moon, Jupiter's Galilean moons	Orbit planets
Asteroids	In the asteroid belt (Mars–Jupiter)	Orbit the Sun; rocky
Comets	Halley's, Hale–Bopp	Very elliptical orbits; ice + dust
Artificial satellites	ISS, GPS, communications	Orbit Earth or other bodies

P6.3 The Life Cycle of Stars

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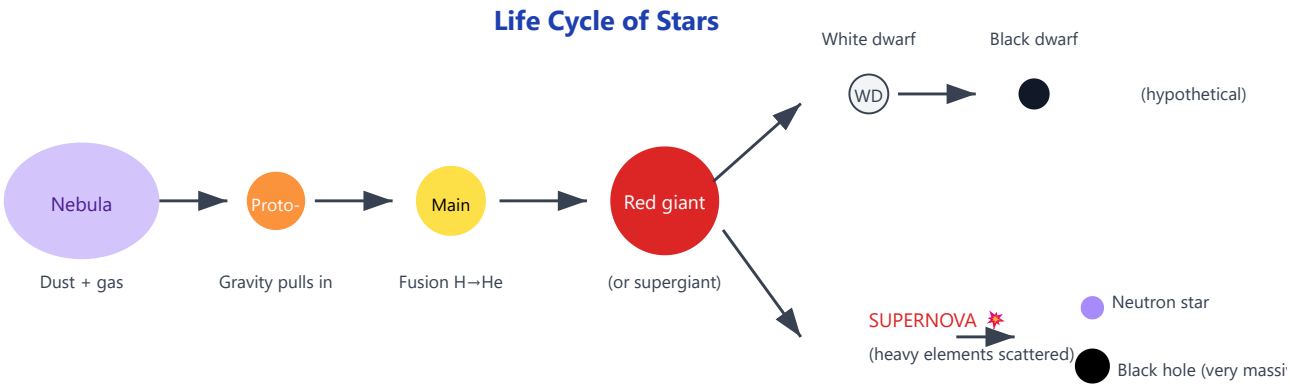


Heavier elements (beyond iron) are forged in supernovae — we are "star dust".

1. **Nebula** — cloud of gas (mostly H) and dust collapses under gravity.
2. **Protostar** — density and temperature rise as matter falls in.
3. **Main sequence star** — temperature high enough for hydrogen fusion (H → He). This releases energy that balances gravitational collapse. Stars spend most of their lives here; the Sun is currently in this stage and has been for ~4.6 Gyr.
4. **Red giant (small/medium star) OR red supergiant (massive star)** — hydrogen runs out, core contracts, outer layers expand and cool. Heavier elements (up to iron) fused.
5. Small/medium star → planetary nebula → **white dwarf** → eventually (if cooled) black dwarf.
6. Massive star → **supernova** explosion → either **neutron star** or (if very massive) **black hole**. Supernovae create elements heavier than iron and scatter them across space.

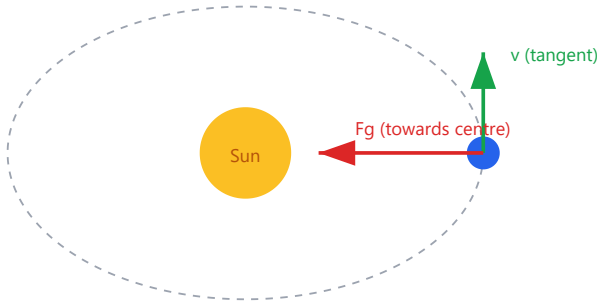
Our Galaxy and Beyond

- Our Sun is one star in the **Milky Way** galaxy, which contains ~200 billion stars.
- The Milky Way is one of billions of galaxies in the **observable universe**.
- Light travels at 3×10^8 m/s. A **light-year** is the distance light travels in one year (~9.5×10^{15} m). Astronomical distances are huge.

P6.4 Orbital Motion

An object moving in a circular orbit has a velocity that is **constantly changing direction**. A change in velocity means acceleration, which requires a resultant force: **gravity** provides this force, pulling the orbiting body towards the centre.

Orbital Motion — Gravity = Centripetal Force



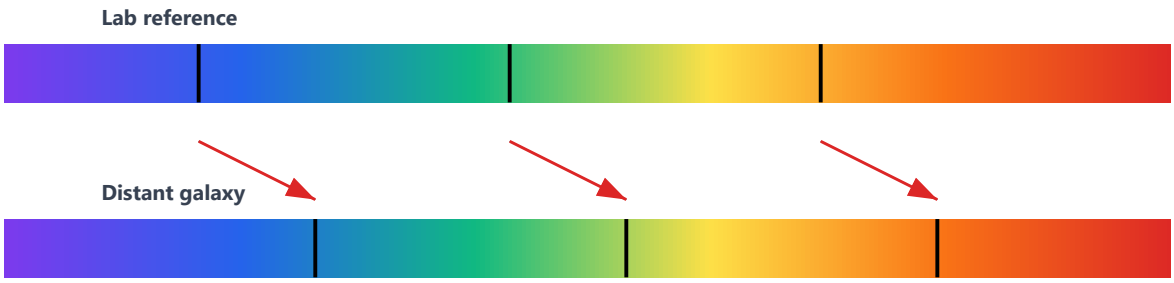
Speed is constant, but velocity (direction) changes. Gravity gives the required centripetal force.

- For a **stable orbit** at a given distance, there is only one specific orbital speed — too fast and the orbit spirals outwards; too slow and it spirals inwards.
- A planet closer to the Sun needs a **higher orbital speed** to balance the stronger gravitational force.
- Satellites in **low Earth orbit** (ISS at ~400 km) travel at ~7.7 km/s; **geostationary satellites** (~36 000 km) at ~3 km/s.

P6.5 Red-Shift and the Big Bang

Red-shift is the observation that light from distant galaxies has a **longer wavelength** than expected (shifted toward the red end of the spectrum). The further away a galaxy is, the greater its red-shift — so it is moving away faster.

Red-Shift — Light from Distant Galaxies Stretched



Dark absorption lines shifted to longer (redder) wavelengths → galaxy moving away from us.

The interpretation: space itself is expanding, carrying galaxies along with it. The universe started very small, hot and dense and has been expanding ever since. This is the **Big Bang** theory (~13.8 billion years ago).

Evidence for the Big Bang

- **Red-shift** of distant galaxies — universe is expanding.
- **Cosmic microwave background radiation (CMB)** — cold microwave glow filling the sky in all directions; the "afterglow" of the hot early universe, redshifted to microwaves by 13.8 Gyr of expansion.
- **Abundance of light elements** (hydrogen, helium) matches Big Bang nucleosynthesis predictions.

Dark Matter and Dark Energy

Observations of galaxy rotation and cosmic expansion cannot be explained by the ordinary matter we can see:

- **Dark matter** — matter that doesn't emit or absorb light but has mass. Its gravitational effect is needed to explain how fast galaxies rotate.
- **Dark energy** — mysterious form of energy driving the **accelerating** expansion of the universe discovered in 1998.
- Current best estimate: ~5% ordinary matter, ~27% dark matter, ~68% dark energy. Physicists do not yet know what either of the "dark" components actually is — an active research area.

P6.6 Remote Sensing & Communications

Artificial satellites are used for:

- **Communications** (TV, phone, internet) — geostationary satellites stay above one point on Earth.
- **Navigation** — GPS uses timing signals from ≥4 satellites to triangulate position.
- **Weather** — monitor cloud movements and ocean temperatures.
- **Earth observation** — deforestation, ice-cap monitoring, climate data.
- **Astronomy** — telescopes above the atmosphere (Hubble, JWST) see wavelengths blocked by air and escape distortion.

Absorption by the Atmosphere

Different wavelengths of the EM spectrum are absorbed by different parts of the atmosphere:

- Most UV is absorbed by the ozone layer.
- Most IR is absorbed by water vapour and CO_2 (greenhouse effect).
- Radio and visible wavelengths pass through the atmosphere easily — "windows" used for ground-based astronomy.

> **Exam Tip:** Why does the JWST orbit far from Earth (L2 Lagrange point)? Because it observes infrared — sitting 1.5 million km away lets it cool to ~40 K, away from the warm Earth and Sun, so its own heat does not overwhelm faint astronomical IR signals.

P6.7 The Future — Global Challenges in Physics

Challenge	Physics role
Climate change	Clean energy (renewables, fusion, nuclear), efficient grid storage, carbon capture
Energy security	Diversify mix; domestic renewables; fusion in the medium-term
Medical imaging	MRI, CT, PET, ultrasound — safer & higher resolution diagnostics
Communications	5G / 6G networks, satellite internet, optical-fibre bandwidth

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Transport	EV batteries, hydrogen fuel cells, maglev, electric aircraft
Space exploration	Planetary science, asteroid defence, space-based solar power

Quick Summary — Key Equations & Ideas for P6

Quantity	Equation / idea	Units
Transformer (voltage)	$V_p/V_s = N_p/N_s$	—
Ideal transformer power	$V_pI_p = V_sI_s$	W
Orbital motion	Gravity acts as centripetal force	—
Speed of light in vacuum	$c = 3 \times 10^8$ m/s	m/s
Red-shift → expansion	Longer λ from distant galaxies	—
Age of universe	$\approx 13.8 \times 10^9$ years	—

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